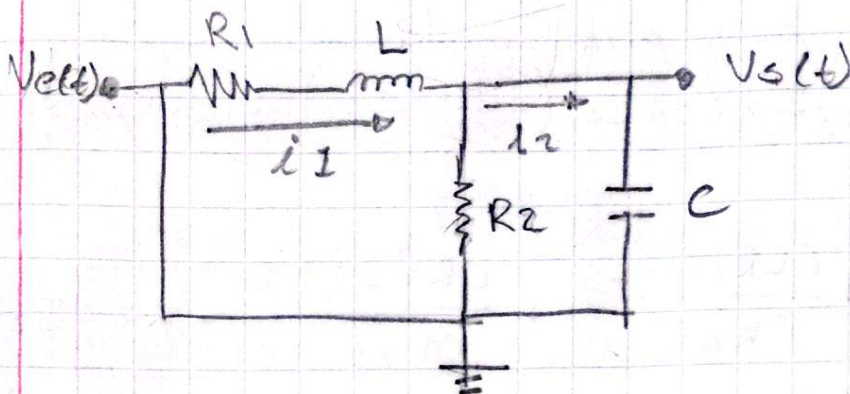


Sistema circulatorio



$$\begin{aligned} R_1 &= 3 \text{ K}\Omega \\ L &= 330 \text{ mH} \\ R_2 &= 300 \text{ K}\Omega \\ C &= 3.3 \text{ mF} \end{aligned}$$

$$\begin{aligned} V_e(t) &= R_1 \dot{i}_1 + L \frac{d\dot{i}_1(t)}{dt} + R_2 [\dot{i}_1(t) - \dot{i}_2(t)] \\ R_2 [\dot{i}_1(t) - \dot{i}_2(t)] &= \frac{1}{C} \int \dot{i}_2(t) dt \\ V_s(t) &= \frac{1}{C} \int \dot{i}_2(t) dt \end{aligned}$$

Transformado de Laplace

$$V_e(s) = R_1 I_1(s) + L s I_1(s) + R_2 [I_1(s) - I_2(s)]$$

$$R_2 [I_1(s) - I_2(s)] = \frac{I_2(s)}{s}$$

$$V_e(s) = I_1(s) (R_1 + Ls + R_2) - R_2 I_2(s)$$

$$V_s(s) = \frac{I_2(s)}{s}$$

$$\frac{V_e(s) + R_2 I_2(s)}{R_1 + Ls + R_2} = I_1(s)$$

Despeje $I_1(s)$

$$R_2 I_1(s) - R_2 I_2(s) = \frac{I_2(s)}{s}$$

$$R_2 I_1(s) = \frac{I_2(s)}{s} + R_2 I_2(s)$$

$$R_2 I_1(s) = \left(\frac{1}{Cs} + R_1 \right) I_2(s)$$

$$I_1(s) = \left(\frac{1}{CR_2 s} + \frac{CR_2 s}{CR_2 s} \right) I_2(s)$$

$$I_1(s) = \left(\frac{CR_2 s + 1}{CR_2 s} \right) I_2(s)$$

Substitución en Ecu. prim. malla

$$V_e(s) = R_1 \left(\frac{CR_2s + 1}{CR_2s} \right) I_2(s) + Ls \left(\frac{CR_2s + 1}{CR_2s} \right) I_2(s) + R_2 \left(\frac{CR_2s + 1}{CR_2s} \right) I_2(s) - R_2 I_2(s)$$

$$V_e(s) = \left[\frac{CR_1R_2s + R_1}{CR_2s} + \frac{CLR_2s^2 + Ls}{CR_2s} + \frac{CR_2^2s + R_2}{CR_2s} - R_2 \right] I_2(s)$$

$$V_e(s) = \left[\frac{CLR_2s^2 + CR_2^2s + CR_1R_2s + Ls + R_1 + R_2}{CR_2s} - R_2 \left(\frac{CR_2s}{CR_2s} \right) I_2(s) \right]$$

$$V_e(s) = \left[\frac{CLR_2s^2 + CR_2^2s + CR_1R_2s + Ls + R_1 + R_2}{CR_2s} - \frac{CR_2^2s}{CR_2s} \right] I_2(s)$$

$$V_e(s) = \left[\frac{CLR_2s^2 + CR_2^2s + CR_1R_2s + Ls + R_1 + R_2}{CR_2s} - \frac{CR_2^2s}{CR_2s} \right] I_2(s)$$

$$V_e(s) = \left[\frac{CLR_2s + (CR_1R_2 + L)s + R_1 + R_2}{CR_2s} - \frac{CR_2^2s}{CR_2s} \right] I_2(s)$$

$$V_e(s) = \left[\frac{CLR_2s^2 + (CR_1R_2 + L)s + R_1 + R_2}{CR_2s} \right] I_2(s)$$

$$\frac{V_s(s)}{V_e(s)} = \frac{R_2}{CLR_2s^2 + (CR_1R_2 + L)s + R_1 + R_2}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{R_2}{LR_2Cs^2 + (L + R_1R_2C)s + (R_1 + R_2)}$$

Primer integral diferencial i_1

$$V_e(s) + R_2 I_2(s) = R_1 I_1(s) + L s I_1(s) + R_2 I_1(s)$$

$$V_e(s) + R_2 I_2 = I_1(s) [R_1 + L s + R_2]$$

$$I_1(s) = \frac{V_e(s) + R_2 I_2(s)}{R_1 + L s + R_2}$$

$$V_e(s) - L s I_1(s) - R_2 I_1(s) + R_2 I_2(s) = R_1 I_1(s)$$

$$I_1(s) = \frac{V_e(s) - L s I_1(s) - R_2 I_1(s) + R_2 I_2(s)}{R_1}$$

Segunda integral diferencial i_2

$$R_2 [i_1 - i_2(t)] = \frac{1}{C} \int i_2(t) dt$$

$$R_2 i_1(t) - R_2 i_2(t) = \frac{1}{C} \int i_2(t) dt$$

$$R_2 i_1(t) - R_2 i_2(t) = \frac{1}{C} \int i_2(t) dt$$

$$R_2 i_1(t) = (R_2 + \frac{1}{C} \int dt) i_2(t)$$

$$i_2(t) = i_1(t) - \frac{1}{R_2 C} \int i_2(t) dt$$

$$\frac{di_1}{dt} = \frac{1}{C} (i_1(t) - i_2(t)) \quad i_2(t) = \frac{V_s(t)}{R_2}$$

$$\frac{dV_s}{dt} = \frac{1}{C} (i_1(t) - \frac{V_s(t)}{R_2})$$

$$\frac{di_1}{dt} = \frac{1}{L} [V_e - R_1 i_1(t) - V_s(t)]$$